

INTERN & PROJECT DETAILS

***COMPANY NAME: CODTECH
IT SOLUTIONS***

INTERN NAME: KAKARLAMUDI DIVYA

INTERN ID: CITS998

DOMAIN: VLSI DESIGN

BATCH DURATION: MAY-JUNE 2026

TASK TITLE: DIGITAL CLOCK CIRCUIT DESIGN AND SIMULATION

PROJECT OVERVIEW & OBJECTIVE

The main objective of this task is to design and simulate a synchronous 24-Hour Digital Clock Circuit using Verilog Hardware Description Language (HDL). The circuit accurately tracks real-time progress in hours, minutes, and seconds. The design is verified using a testbench environment simulated on EDA Playground.

KRY FEATURES:

24-Hour Format: Tracks time continuously from 00:00:00 up to 23:59:59.

Synchronous Reset: Instantly resets the complete clock registers back to 00:00:00 when the reset (rst) signal is high on a clock edge.

Hierarchical Counting: Seconds increment minutes upon reaching 59, and minutes increment hours upon reaching 59.

PIN CONFIGURATION & DESCRIPTION

<i>SIGNAL NAME</i>	<i>DIRECTIO N</i>	<i>BIT WIDTH</i>	<i>DISCRIPTI ON</i>
<i>Clk</i>	<i>Input</i>	<i>1-bit</i>	<i>Global System Clock Signal used for synchroni zation</i>
<i>Rst</i>	<i>Input</i>	<i>1-bit</i>	<i>Active-Hig h Synchron</i>

			<i>ous Reset to clear clock registers to 00:00:00.</i>
<i>Sec</i>	<i>Output</i>	<i>6-bit</i>	<i>Output register to track seconds (Counts from 0 to 59).</i>
<i>Min</i>	<i>Output</i>	<i>6-bit</i>	<i>Output register to track minutes (Counts from 0 to 59).</i>
<i>Hrs</i>	<i>Output</i>	<i>5-bit</i>	<i>Output</i>

			<i>register to track hours (Counts from 0 to 23).</i>
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SOURCE CODE

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// COMPANY : CODTECH IT SOLUTIONS

// INTERN NAME : KAKARLAMUDI DIVYA

// INTERN ID : CITS998

// DOMAIN : VLSI DESIGN ENGINEER

// TASK TITLE : DIGITAL CLOCK CIRCUIT

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VERILOG DESIGN CODE

module digital_clock (

input wire clk, // Main system clock input

input wire rst, // Active-high synchronous

reset signal

**output reg [5:0] sec, // 6-bit register to store
seconds (0 to 59)**

output reg [5:0] min, // 6-bit register to store minutes (0 to 59)

output reg [4:0] hrs // 5-bit register to store hours (0 to 23)

);

// Sequential logic triggered at the rising edge of the clock

always @(posedge clk) begin

if (rst) begin

// Synchronous Reset: Clear all time registers to 00:00:00

sec <= 6'd0;

min <= 6'd0;

hrs <= 5'd0;

end else begin

// Check if seconds reached upper limit

(59)

if (sec == 6'd59) begin

***sec <= 6'd0; // Reset seconds counter
to 0***

***// Check if minutes reached upper
limit (59)***

***if (min == 6'd59) begin
min <= 6'd0; // Reset minutes
counter to 0***

***// Check if hours reached upper
limit of 24-hour format (23)***

***if (hrs == 5'd23) begin
hrs <= 5'd0; // Reset hours
counter to 0 (New Day starts)
end else begin
hrs <= hrs + 1'b1; // Increment
hours counter by 1
end
end else begin***

```
        min <= min + 1'b1; // Increment
minutes counter by 1
    end
    end else begin
        sec <= sec + 1'b1; // Increment
seconds counter by 1
    end
end
end
endmodule
```

TEST BENCH CODE

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```
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// TASK TITLE   : DIGITAL CIOCK CIRCUIT  
//
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TEST BENCH CODE:

```
`timescale 1ns / 1ps
```

```
module tb_digital_clock;
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```
    // Testbench inputs declared as 'reg' to drive  
stimulus
```

```
    reg clk;
```

```
    reg rst;
```

***// Testbench outputs declared as 'wire' to
monitor signals***

wire [5:0] sec;

wire [5:0] min;

wire [4:0] hrs;

// Instantiate the Unit Under Test (UUT)

digital_clock uut (

.clk(clk),

.rst(rst),

.sec(sec),

.min(min),

.hrs(hrs)

);

***// Clock Generator: Toggle clock every 5ns
(Creates a 100MHz clock)***

always begin

#5 clk = ~clk;

end

```

// Stimulus Block
initial begin
    // Initialize inputs
    clk = 0;
    rst = 1; // Assert reset at startup

    // Hold reset condition for 20ns
    #20;
    rst = 0; // Deassert reset to let the clock
run

    // Monitor signal changes automatically in
the EDA log terminal
    $monitor("Time = %0t |
%02d:%02d:%02d", $time, hrs, min, sec);

    // Run simulation for a long duration to
observe increments
    #150000;

```

```
// Terminate simulation run  
$finish;  
end  
  
// Waveform Configuration Block for EDA  
Playground EPWave display  
initial begin  
    $dumpfile("dump.vcd");           // Specifies  
the VCD waveform file name  
    $dumpvars(1, tb_digital_clock); // Dumps  
top-level testbench variables  
end  
  
endmodule
```

SIMULATION RESULTS & OUTPUTS

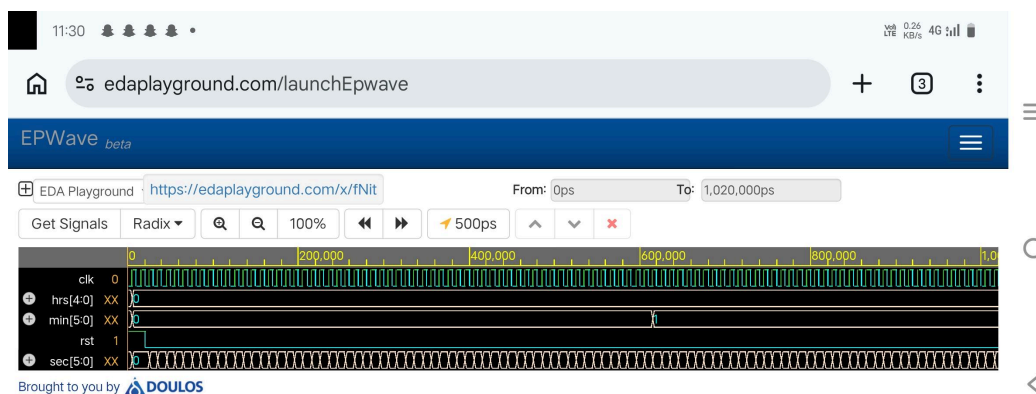
The functional correctness of the 24-Hour Digital Clock design is verified by executing the testbench on the Aldec Riviera-PRO simulator. The resulting console output and timing waveforms demonstrate that the clock counts correctly.

A. Consol output log:

The console output log captures the real-time values of hours, minutes, and seconds as they increment sequentially on every clock pulse.

. Timing Waveform (EPWave Window:

The EPWave diagram shows the graphical representation of signals including clk, rst, sec, min, and hrs. It illustrates how the counters increment on the rising edge of the clock.



CONCLUSION:

The 24-Hour Digital Clock Circuit was successfully designed using Verilog HDL and verified using the Aldec Riviera-PRO simulation engine on the EDA Playground platform. The behavioral testbench effectively applied stimuli to evaluate both the functional paths and the synchronous reset mechanism. The resulting console text logs and graphical timing waveforms confirmed that all conditions for time transitions—specifically the rolling over of seconds to minutes and minutes to hours—function correctly without any runtime errors. This successfully fulfills all the technical requirements specified for this internship task by Codtech IT Solutions.

